Novasphere Dynamics - Autonomous Robotics Safety & Deployment Brief

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1. Purpose & Scope

This technical brief specifies mandatory field safety protocols for deploying autonomous mobile robots (AMRs) and industrial drones connected to the Novasphere OS control plane.

2. Physical Safety Perimeter & E-Stop Requirements

During active commissioning, all human personnel must maintain a minimum 3-meter physical safety perimeter around active AMR testing zones. All robotic units must be outfitted with a dual-channel hardware Emergency Stop (E-Stop) button rated at SIL-3 / PLe. In the event of network disconnection exceeding 40ms, ZeroGravity failover or automatic safe-braking will engage.

3. Radio Frequency & Telemetry Prerequisites

Industrial deployment facilities must provide dedicated Wi-Fi 7 or private 5G coverage on UDP port 9872. Minimum signal strength must exceed -65 dBm across all operating corridors. If signal drops below -75 dBm, the node must decelerate to a crawl speed of 0.2 m/s.

4. Incident Reporting & Escalation

Any safety near-miss or hardware anomaly must be logged within 60 minutes via the internal Safety Portal. For critical Severity-1 emergencies involving physical damage, contact the 24/7 Operations Hotline immediately at +41 44 888 0199.